Your guide to OSTEOSARCOMA

BONE CANCER RESEARCH TRUST

UNTIL THERE'S A CURE

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JUST DIAGNOSED WITH BONE CANCER?

Hearing that you have cancer is scary and there will be lots to take in. We feel that it's important for you to understand your diagnosis in order to help you feel a better sense of control and understand more about what is happening to you.

This booklet is designed to explain what osteosarcoma is, what treatments you will receive, how they may affect you, how to help with side-effects, how you may feel and what happens after your treatment.

It's easy to think that only old people get cancer because over 65% of cancer is found in people over the age of 65. But some types of cancer mainly affect children, teenagers and young people. Primary bone cancer is one of these.

Bone cancer develops when a cell within the bone starts to grow abnormally

Cancer is a disease OF THE BODY'S cells

CELLS

OKAY, BUT WHAT IS A CELL?

All living things are made up of cells. Cells are the building blocks for tissues and organs that make up our bodies. Each human being is thought to be made of more than 37 trillion cells (that's 37,000,000,000,000 cells!).

37 TRILLION CELLS

During development, growth and when the body needs to repair itself, new cells need to be made. New cells are made when one cell splits into two to form two new exact copies; which are called daughter cells. We call this 'cell division.'

The process of cell division is very tightly controlled; normal cells know when to start and when to stop dividing, and when it is time for the cell to die. This means new cells are only made when they are needed. Each cell has a set of instructions that tells it what type of cell it is and when it should divide. We call these instructions 'genes'.

If cells start to divide too quickly or in an uncontrolled way, they can form a lump called a tumour. As well as dividing uncontrollably, these cells can also invade and damage other tissues, or break away from the tumour and spread to other parts of the body and cause damage there. When this happens, the lump or tumour is called 'malignant' and we call this condition 'cancer'.

WHY DO THESE CELLS START TO DIVIDE TOO QUICKLY?

Certain genes are responsible for telling the cell when to start and stop dividing and when it is time for the cell to die. If these genes are damaged, then rather than dying, those cells can continue to live and divide; making more and more cells.

CANCER IS NOT A SINGLE DISEASE, THERE ARE OVER 200 DIFFERENT TYPES AFFECTING MANY DIFFERENT ORGANS AND TISSUES.

Primary bone cancer is the name given to cancer that starts in the bones, rather than cancer that has spread to the bones from other parts of the body, which is called secondary bone cancer.

BONES

Did You know?
BONE SARCOMA IS ANOTHER TERM FOR BONE CANCER

THE DOCTOR SAID BONE CANCER IS VERY RARE, BUT HOW RARE?

Primary bone cancers, also known as 'bone sarcomas', are rare types of cancer. Each year around 600 people in the UK and the Republic of Ireland will be told they have a primary bone cancer and around a third of these will be children, teenagers or young adults.
Over one third of cases are seen in people under the age of 29 – so you're definitely not alone in this.

BONES FEEL LIKE STONE TO TOUCH, HOW CAN THEY GET CANCER?

Although bones appear to be hard and rock-like, they are living structures; they contain cells, nerves and blood vessels just like other parts of the body. The bones in your body are different from what you may see in a skeleton in a school lab or in a museum. They are constantly changing while you grow and as your body changes throughout your life.

As a baby, your bones were very soft and flexible and made from a tissue called cartilage. During growth this cartilage was replaced by hard bone, and so the bones became more solid and rigid. A type of bone cell called an osteoblast makes the bony material and replaces the cartilage. Another type of bone cell called an osteoclast eats away at the bone taking away any which isn't needed. The two types of cells help to shape and repair the bones. By the time you are around 25 years old this process of making your skeleton will be complete. Throughout adult life new bone is made to replace old and damaged bone.

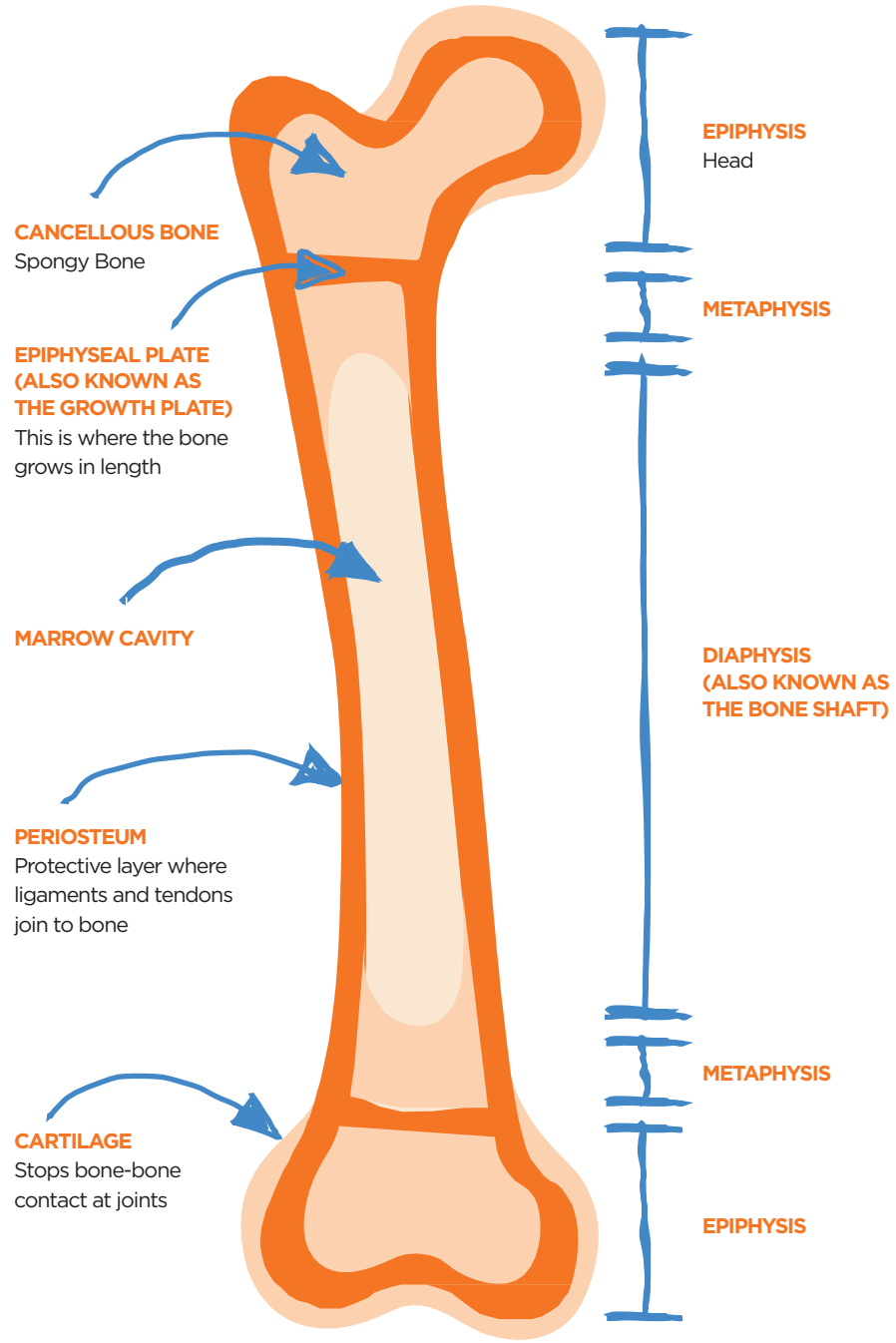
Your bones grow in the same way as the rest of your body; by cells dividing. Near the ends of bones there is an area called the growth plate, it is this part of the bone that gets longer and where lots of cell division takes place. These cells that divide rapidly when your bones grow longer, may be where primary bone cancer starts.

The hollow centre of some bones is the place where bone marrow is made. Bone marrow produces blood cells. People can get cancers which start in the bone marrow but these types of cancer, such as myeloma, leukaemia and lymphoma are not classified as primary bone cancers by doctors.

If you need information on these types of cancer please visit:
www.bloodwise.org.uk

Bone functions

- PROVIDE SHAPE AND SUPPORT
- PROTECT BODY STRUCTURES
- STORE MINERALS
- AID BODY MOVEMENT



TYPES OF BONE CANCER

MY FRIEND ON THE WARD HAS BONE CANCER BUT IT'S NOT THE SAME AS MINE, IS THERE MORE THAN ONE TYPE?

Yes, there are 7 different types of primary bone cancer in total and the 2 most common types occurring in young people are:

- **OSTEOSARCOMA**
- **EWING SARCOMA**

The Bone Cancer Research Trust provides in-depth information on all types of primary bone cancer at **www.bcrt.org.uk/information**, or you can ask your nurse for more information materials.

Did you know?

'OSTEO' COMES FROM AN ANCIENT GREEK WORD FOR BONE AND A 'SARCOMA' IS THE NAME GIVEN TO CANCERS THAT START IN CONNECTIVE OR SUPPORTING TISSUES (SUCH AS BONE, FAT, CARTILAGE, BLOOD VESSELS AND MUSCLE).

TO FIND OUT MORE VISIT WWW.BCRT.ORG.UK/OSTEOSARCOMA

I'VE BEEN TOLD I HAVE OSTEOSARCOMA, I'VE NEVER HEARD OF IT. WHAT IS IT?

Osteosarcoma is the most common type of primary bone cancer in children, teenagers and young adults. It starts when the genes in a bone cell become abnormal, causing the cell to divide out of control.

The cancer cells in the tumour started out from one bone cell, and often produce bony material as they grow. Therefore, when doctors look at a sample of the tumour under a microscope, if they can see this abnormal bone it helps them to confirm a diagnosis of osteosarcoma.

Most osteosarcomas develop in a single place within the area of a long bone, at the site of the growing areas of the bone; which is called the growth plate or epiphyseal plate. A long bone is very much longer than it is wide. The best examples are the thigh bone (femur), the shin bone (tibia) or the upper arm bone (humerus).

- **In the UK and the Republic of Ireland around 175 people are diagnosed with osteosarcoma every year.**
- **Osteosarcoma affects slightly more boys than girls.**
- **Around 60% of osteosarcoma cases are diagnosed in people under the age of 29 years of age.**

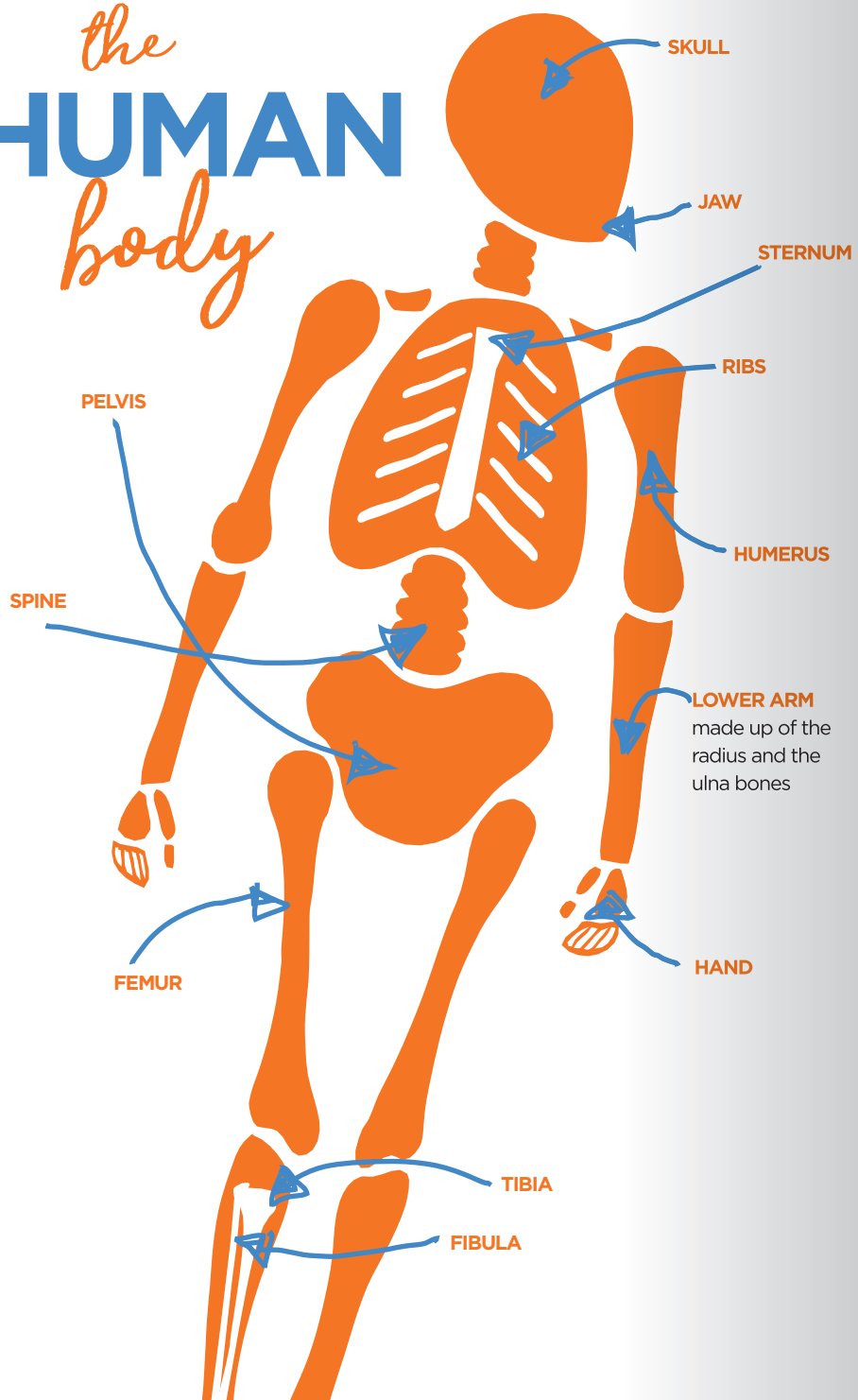
THE TWO MOST COMMON TYPES OF

BONE CANCER

in young people

ARE OSTEO SARCOMA *and* EWING SARCOMA

the HUMAN body



I HAVE AN OSTEOSARCOMA NEAR MY KNEE, WHERE ELSE CAN IT OCCUR?

Most osteosarcomas are found in the lower thigh bone nearest to the knee, the upper shin bone nearest to the knee or the upper arm bone nearest to the shoulder.

Other bones can be affected such as the jaw, spine and the pelvis. Tumours found in the bones of the face, skull and spine are more common in older patients.

ANOTHER PATIENT ON MY WARD HAS OSTEOSARCOMA BUT THE TREATMENT IS DIFFERENT TO MINE, WHY?

There is more than one type of osteosarcoma and the treatment you will receive depends on what type you have.

There are **five** main types of osteosarcoma are:

OSTEOBLASTIC OSTEOSARCOMA

In most cases of osteosarcoma the cancer cells look like bone-forming cells. These cells are called osteoblasts, and therefore this type of osteosarcoma is known as osteoblastic osteosarcoma.

CHONDROBLASTIC OSTEOSARCOMA

In some cases of osteosarcoma the cells look different under the microscope and the cancer cells resemble cartilage-forming cells and bone. Cartilage is tissue that covers and cushions the ends of bones, and the cells that make cartilage are called chondroblasts. Therefore this kind of tumour is called chondroblastic osteosarcoma.

FIBROBLASTIC OSTEOSARCOMA

If the cancer cells are making only a small amount of bone, and the cells look like a kind of cell called a 'fibroblast', then the tumour is called fibroblastic osteosarcoma.

TELANGIECTATIC OSTEOSARCOMA

The word 'telangiectatic' means that there are lots of blood vessels. If lots of abnormal blood vessels are visible in the tumour as well as the bone-forming cells, then the tumour is called telangiectatic osteosarcoma.

SMALL CELL OSTEOSARCOMA

If the tumour is made of small round cells then it is called 'small cell osteosarcoma'. These can sometimes be difficult to tell from other tumours that also contain small round cells.

Most osteosarcomas will be one of these five types and will start inside the bone; these types are 'high grade' tumours. High grade means that when the cells of the tumour are looked at under a microscope many cells are in the process of dividing to make new cancer cells. This makes the tumour behave in an aggressive manner, meaning that the cancer cells can sometimes spread to other parts of the body.

PAROSTEAL AND PERIOSTEAL OSTEOSARCOMA

Sometimes, osteosarcoma can start on or close to the surface of bone. These kinds of osteosarcoma can be less aggressive and are described as 'low grade' and so they are treated differently. These may be called 'parosteal' or 'periosteal' osteosarcoma.

EXTRAOSSEOUS OSTEOSARCOMA

Very rarely, osteosarcoma tumours can grow outside of the bones in the soft tissues of the body. This is known as 'extraosseous osteosarcoma', which means 'osteosarcoma that is outside of the bones'. Extraosseous osteosarcoma is treated in the same way as osteoblastic osteosarcoma.

TREATMENT

HOW DID I GET OSTEOSARCOMA?

Doctors don't really know what causes most cases of osteosarcoma. There is a lot of research going on to try to understand why some people develop bone cancer such as osteosarcoma.

We do know that:

- **You can't catch osteosarcoma from another person**
- **It's not caused by something that you have done, and we don't know of any way to prevent it**

During teenage or early adult years, bones grow very quickly. It is thought that this phase of rapid bone growth may be part of the reason why young people develop osteosarcoma more often during teenage years than in younger children and older adults whose bones grow more slowly.

Risk factors are used to describe the different things that can increase the chance of getting an illness. There are a few known risk factors for osteosarcoma, but these have a very slight effect and are not preventable.

HOW WILL MY OSTEOSARCOMA BE TREATED?

If doctors think you may have bone cancer the type of treatment you will receive will depend on the answer to two questions:

- **What is it?**
- **Where is it?**

WHAT IS IT?

X-rays and scans may show that you are very likely to have osteosarcoma but doctors cannot be fully certain without taking a small piece of the suspected tumour during a test known as a biopsy. The biopsy goes to a special laboratory so that cells from it can be examined under a microscope.

By looking at the cells in the laboratory, an expert doctor called a pathologist can tell whether you have osteosarcoma. The pathologist can also find out what kind of osteosarcoma you have.

Risk factors

Age: most osteosarcoma cases are recording in patients between 10 and 24 years of age

Height: You have a slightly higher risk of getting osteosarcoma if you are tall for your age

Gender: osteosarcoma is slightly more common in boys than in girls

WHERE IS IT?

X-rays and scans also help doctors to see the size of the tumour and exactly where in the body it is. The scans also check whether the cancer has spread to any other parts of the body. This is known as 'staging' of the cancer.

If a cancer has spread away from the original (primary) site it is called 'secondary cancer' and doctors call these secondary tumours 'metastases'.

Osteosarcoma can sometimes spread to other parts of the body and does so to the lungs more often than anywhere else; this is known as 'pulmonary metastases'. Occasionally osteosarcoma can spread to other bones or more unusual sites.

WHAT? MORE TESTS! WHY?

As well as blood tests, you will have tests to check how well your heart, kidneys and liver are working.

This is because some of the medicines used to treat osteosarcoma can affect these organs, so doctors need to know how well they are working before treatment starts.

WHERE WILL I BE TREATED?

You will be treated at a Bone Cancer Centre, which are specialist centres that have a group of healthcare specialists who are experts in diagnosing and treating primary bone cancer. You may have already been seen at a Bone Cancer Centre for your biopsy and tests during diagnosis.

In England, there are currently five Bone Cancer Centres where the surgical treatment of bone cancer is carried out (see these on the map on page 14). These are at Birmingham, Newcastle, Oswestry, Oxford and Stanmore (North London).

You may have to travel a long way to get to a Bone Cancer Centre. If you live in Wales you may go to Oswestry or Birmingham for surgical treatment. In Scotland, you will be seen in one of three centres in Glasgow, Edinburgh or Aberdeen. In Northern Ireland, patients are seen in Belfast. In the Republic of Ireland, there are no specific Bone Cancer Centres; you will probably be treated at a specialist hospital in Dublin.

If you would like further information or have any questions regarding osteosarcoma, please get in touch with the Bone Cancer Research Trust at

0113 258 5934

or by emailing

info@bcrt.org.uk

In-depth information can also be found on our website at

www.bcrt.org.uk/osteosarcoma

Other treatments such as chemotherapy and radiotherapy are often given in hospitals closer to home; this will be arranged by the Bone Cancer Centre and is known as a 'Shared Care' arrangement.

WHO WILL I BE TREATED BY?

You will be treated by a team of people who are experts in bone cancer. These people will include doctors, surgeons, nurses, social workers, dieticians, physiotherapists and many other experts.

This group of people are called a Multidisciplinary Team; you will probably hear it being called the 'MDT'. The MDT will plan and decide on what treatment and care is best for you. But don't worry, they will include you in the decision making.

WHAT ARE THE TREATMENTS FOR OSTEOSARCOMA?

Treatments for osteosarcoma typically includes chemotherapy (often called chemo), surgery to remove the tumour and, on some occasions, radiotherapy.

CHEMOTHERAPY

Chemo is able to travel around the whole body through the bloodstream. It is used to shrink the tumour before surgery and kill any cancer cells that could have spread from the original (primary) tumour site.

In most countries, chemo is given both before and after surgery. After you have been told you have osteosarcoma it is likely that your chemo will begin quickly; but the doctors will talk to you and answer any questions you may have about your treatment plan first.

CHEMOTHERAPY BEFORE SURGERY (NEO-ADJUVANT CHEMOTHERAPY)

Chemo given before surgery is called neo-adjuvant chemotherapy. Usually, the chemo given before surgery is made up of 3 drugs given over 10 weeks. The aim of this course of chemo is to make the tumour smaller before surgery, and to kill any cells that may have spread to other parts of the body.

SURGERY

Following these cycles of chemo, the next part of your care is to remove the main tumour using surgery. The MDT team will decide on what type of surgery may be best for you. But don't worry, you will also be part of the decision making process.

The aim of surgery is to remove the primary tumour safely and at the same time try to keep the body working as normally as possible. If the primary tumour is in a limb then limb-sparing surgery may be possible.

There are many different 'limb-sparing' techniques that surgeons use to preserve limbs, the main one being replacement of the affected bone with a metal implant and false joint. Another technique is to carry out an autologous bone graft. This is where healthy bone is taken from a different part of the body to replace the bone damaged by the tumour.

Even with these advances in surgery around 15% of patients require an amputation (removal of the limb) to safely remove the tumour. Patients who have amputations can go on to lead independent, active and full lives, sometimes having added inspiration to achieve great things.

When the tumour is surgically removed, some healthy tissue surrounding the tumour will also be taken. This is then examined under a microscope to check that the entire tumour has all been removed. The result of this examination may affect the treatment a patient receives after surgery.

CHEMO AFTER SURGERY (ADJUVANT CHEMOTHERAPY)

Following the removal of the tumour, some patients go on to receive further chemo; this is known as adjuvant chemo.

If there is evidence that the tumour has spread to other parts of the body then an oncologist (a doctor who cares for cancer patients) and a surgeon may want to think about the possibility of removing the secondary tumours by surgery.

Your oncologist is the best person to describe treatment choices with you and tell you what to expect from the treatment. Treatment of cancer involves patients and the doctors working together to find a care or treatment plan that fits their needs.

You can read more details about how chemo works, how it is given and the possible side-effects in the 'Chemotherapy' section of this booklet on page 24.

RADIOTHERAPY

Radiotherapy means treating the tumour with high doses of concentrated radiation. Osteosarcomas are not very sensitive to radiation, and so radiotherapy treatment isn't used as often as chemotherapy and surgery.

However, radiotherapy is used on some occasions, such as after surgery, to ensure the entire tumour is completely removed. It can also be used instead of surgery where surgical removal of the osteosarcoma is not possible due to its location.

The main treatments are:

Chemotherapy

Surgery

200

young people

are told

THEY HAVE BONE CANCER

EVERY YEAR IN THE UK

ASKING THE RIGHT QUESTIONS

Being diagnosed with osteosarcoma can be a stressful and overwhelming time. It is often difficult to gain all the information you would like to whilst speaking with your medical team and it is certainly not uncommon to leave an appointment with some unanswered questions.

Here is a list of just a few of the questions you may wish to ask your medical team during your diagnosis or treatment. Think of these as a starting point and don't be afraid to ask any other questions you may have. If you don't understand something, it's OK to ask your doctor or nurse to repeat themselves until you do, or you could ask them to try explaining it in a different way.

QUESTIONS DURING DIAGNOSIS

- Do you know the size and location of my cancer?
- How can I manage my symptoms?
- Will I need to make any changes to my day-to-day life, diet or exercise?
- Who can I contact if I have any further questions, and will I be able to do so out of office hours?
- Who can I talk to if I am feeling overwhelmed with my diagnosis?

QUESTIONS DURING TREATMENT

- What treatment will I receive and how was this treatment plan decided?
- How long will my treatment last?
- How long will I have to stay in hospital?
- What are the likely side-effects to this treatment and will I need medication to control these side-effects?
- Am I at risk of any effects later on in my life, and if so, is there anything I can do to prevent these before my treatment begins?
- Will this treatment affect my day-to-day life, such as my ability to attend work or school?
- What happens if my treatment doesn't work?
- Are there any clinical trials that I could take part in?

QUESTIONS AFTER TREATMENT

- Will I need rehabilitation treatment, and if so, how will it be arranged and where will it take place?
- Will I need follow-up tests to ensure the treatment has worked?
- What is available for me in terms of prescriptions or financial help?
- What support is available for me? For example at school or at work.

Handy Tips

NOTE DOWN ANY QUESTIONS, CONCERNS OR SYMPTOMS YOU WISH TO DISCUSS BEFORE YOUR APPOINTMENTS SO YOU REMEMBER EVERYTHING YOU WANTED TO MENTION.

WRITE IT DOWN! BETTER STILL, HAVE A FAMILY MEMBER OR FRIEND WITH YOU WHO CAN DO THIS FOR YOU WHILE YOU SPEAK WITH THE DOCTOR.

TAKE DOWN THE CONTACT DETAILS OF MEMBERS OF THE TEAM THAT YOU MAY WISH TO SPEAK TO AT A LATER DATE.

SURGERY

I'M HAVING LIMB-SPARING SURGERY, WHAT DOES THAT MEAN?

Most patients with osteosarcoma will undergo surgery to remove their tumour.

If the tumour is in your arm or leg, limb-sparing surgery will usually be carried out. This is surgery where the tumour is entirely removed, along with some surrounding bone and muscle, and a special implant is put into the limb to replace the bone and muscle that has been taken out. The implant allows your arm or leg to work as normally as possible after the operation. The implant is called an endoprothesis. Endo means 'inside' and 'prosthesis' is the name given to an artificial body part.

Alternatively, some of the bone may be replaced by a bone graft. This means a piece of your own bone from elsewhere in your body will be taken and used to replace the bone that had the tumour in it.

Your implant may even be a complete joint, for example, a knee, hip or shoulder. Although you will be able to do most things as normally as possible, you may still have a limp or find it difficult to run. Your surgeon may advise you not to run or take part in any rough contact sports.

The surgeon will explain to you and your family what type of surgery is best for you. You will be given lots of opportunities to ask as many questions as you like. There's no such thing as a daft or silly question, so if something's worrying you or there is something you don't fully understand, just ask.

Before the operation, the surgeon and a doctor called an anaesthetist will talk to you and explain what will happen during the operation. The anaesthetist is a doctor who specialises in putting patients to sleep or numbing an area of their body for an operation.

REHAB FOLLOWING LIMB-SPARING SURGERY

A physiotherapist will visit you to talk about what will happen before and after the operation, and about rehabilitation (usually shortened to rehab). Rehab is the process of strengthening your body after the operation and learning to live with the changes to your body. The physiotherapist will give you some exercises to do every day before the operation can take place. You may still be a little bit weak after your chemo and the exercises will help you to strengthen your muscles.

You will start rehab very soon after the operation. This will involve physiotherapists helping you to strengthen your muscles by getting you to move the limb that has been operated on and to do lots of exercises to strengthen the limb. You will have regular check-ups with your surgeon to check that the implant is working well, is not damaged and there is no sign of infection.

If you have limb-sparing surgery on your leg, you may have to undergo a procedure called leg lengthening, this is especially important if you are still growing. This is normally done by a special machine in the hospital; it takes just a few minutes and does not hurt. Your surgeon will explain this procedure to you.

THE DOCTOR SAYS I CAN'T HAVE LIMB-SPARING SURGERY, WHY?

It is usually possible to remove a bone tumour from an arm or leg using limb-sparing surgery. However, for around 15% of patients who have a bone tumour in their arm or leg this is not possible.

In these 15% of patients, the tumour may have grown or spread into the main blood vessels or nerves of an arm or leg and this means limb-sparing surgery is not possible because these important structures would need to be removed with the tumour. This means that an amputation will be needed as part of the treatment for bone cancer. An amputation is the removal of all or part of the arm or leg.

If the MDT decides that an amputation is the best treatment for you, your surgeon will tell you why and give you lots of information about what will happen before, during and after the operation.

It may be difficult at first for you to come to terms with the thought of losing part of, or all of an arm or leg. You may have lots of questions and feel emotions such as anger, shock or be feeling scared. You can talk to a counsellor or even other patients who have had similar operations which may help you. If you would like to talk to a counsellor have a word with your doctor or clinical nurse, or ask a family member or friend to ask them.

If there's anything you don't understand, don't be afraid to ask about it again, the doctors and nurses will be happy to go through anything you're unsure of.

Many people have amputations for different reasons and go on to live an independent, active and full life. There will be challenges but with all the help you will receive there's usually no reason why you can't beat them. After amputation, some people are inspired to do lots of great things, like taking part in sports.

I'M HAVING AN AMPUTATION. HOW MUCH OF MY ARM OR LEG WILL I LOSE?

Again, this depends on the question 'where is the tumour?' Each case is different so your surgeon or nurse will explain to you the kind of amputation you will need.

WHAT WILL HAPPEN BEFORE AND DURING MY AMPUTATION SURGERY?

The surgeon, and possibly a physiotherapist, will visit you before the operation to talk about what will happen during and after the operation and about rehab. It is often helpful to visit a Limb Fitting Centre before surgery to talk about rehab, but this is not always possible. At a Limb Fitting Centre you will be able to talk about your care after the operation and go through the options for having an artificial limb (or prosthesis) fitted.

An anaesthetist will also visit you before the operation. During the operation you will be asleep under general anaesthetic. Alternatively, an epidural may be used to completely numb the lower half of the body. With either kind of anaesthetic you will not feel pain during the operation.

WHAT WILL HAPPEN AFTER MY AMPUTATION SURGERY?

When you wake up, you may find that a drain has been put into the remaining part of your limb, which is called the stump (or sometimes referred to as a 'residuum').

This drain helps to take away any fluid that may build up after surgery. A dressing and special stocking will be placed over the stump. It is common for patients to still feel the limb after it has been removed. Occasionally these sensations are painful but the doctors can help with this. You may also receive antibiotics to reduce the risk of infection after the operation.

REHAB FOLLOWING AN AMPUTATION

You will be allowed to rest and recover from the operation for a short period before you begin rehabilitation (also called rehab).

Rehab is the process of strengthening your body after the operation and learning to live with the changes to your body. Rehab may start after as little as 24 hours after surgery. Physiotherapists and occupational therapists are usually involved. Occupational therapists help people who have been ill or have undergone surgery to find ways to adapt to life with the changes caused by their illness; such as helping the patient to get around in day-to-day life or to return to work or school.

At the start of rehab you will be given advice and an exercise plan to improve your strength, fitness and how you move around.

- **Getting fitter is very important because walking with an artificial limb requires more energy than walking with a normal leg.**
- **Advice will be given about the care of the stump.**
- **If you have a leg amputation you will be shown how to move from a bed or a chair to a wheelchair.**
- **There will also be special exercises to get you ready for using an artificial limb.**
- **As the wound heals a temporary prosthesis may be an option for some patients.**
- **A Home Exercise Programme will be explained to you and your family.**

The majority of patients who have had an amputation are able to make use of a prosthetic limb but successful rehabilitation will depend on your motivation, fitness and the condition of the stump. You may have some days when you feel you can't be bothered with it all but you will be given lots of encouragement and help.

When the stump has fully healed you will be able to visit a Limb Fitting Centre. You will be under the care of an MDT consisting of doctors, specialist nurses, physiotherapists, prosthetists (people who specialise in caring for people with artificial limbs) and occupational therapists. The MDT at the Limb Fitting Centre will:

- **Give you the details of your rehab**
- **Help to decide the best artificial limb that meets your needs**
- **Help with the fitting of the limb**
- **Teach you to walk again using the limb and advise you how to get the most out of it**
- **Show you how to care for your artificial limb and the stump**

The specialist nurses at the Limb Fitting Centre and hospital can help you and your family to deal with the emotional aspects of having an amputation, which may include shock, anxiety and fear about the future. With the help and guidance of the MDT you should achieve full independence and good function.

I KEEP GETTING INFECTIONS AND OTHER SET-BACKS, I'M WORRIED MY TREATMENT ISN'T WORKING.

Setbacks are normal. During your treatment, things may not always go as smoothly as the doctors and you would like. There may be times when you feel really unwell or get an infection or may have to have more surgery, but try not to worry, all of these things are a normal part of cancer treatment. It doesn't mean that your treatment isn't working or isn't working as well as was expected. You may wonder why your friend on the ward doesn't get infections or the same side-effects as you but just remember; everyone is different.

During your treatment if you feel up to it, try to arrange a day with your friends or a special treat - have some fun, you are allowed. Talk to your friends about your cancer, maybe even show them this booklet but tell them although you have bone cancer, 'you're still you.'

VISIT THE BONE CANCER RESEARCH TRUST WEBSITE TO READ ABOUT FORMER BONE CANCER PATIENT, **KELLY CUMMING**, WHO WITH HER PROSTHESIS HAS RUN MANY MARATHONS AND LEADS A NORMAL, ACTIVE AND HAPPY LIFE.
WWW.BCRT.ORG.UK/PATIENTSTORIES

CHEMOTHERAPY

HOW DOES THE CHEMO KNOW WHICH CELLS ARE THE CANCER CELLS?

Cancer cells divide and grow quickly, so chemo drugs were developed to target rapidly dividing cells. Different chemo drugs affect different parts of cells and that is why more than one drug may be given. This is called combination chemo.

VISIT THE BONE CANCER RESEARCH TRUST WEBSITE TO HEAR **JAMIE WOOD'S** TIPS ON HOW HE DEALS WITH THE SIDE-EFFECTS OF CHEMO. JAMIE WAS DIAGNOSED WITH BONE CANCER WHEN HE WAS 18 YEARS OLD.
WWW.BCRT.ORG.UK/PATIENTSTORIES

WHAT CAUSES SIDE-EFFECTS AND WHAT WILL THEY BE?

Most healthy cells are not dividing quickly. However, some cell types do divide rapidly, such as hair follicle cells, skin cells, bone marrow cells, and those lining the digestive sy . This means chemo can also affect these healthy cells and this is what causes side-effects.

Side-effects can be unpleasant, such as nausea, diarrhoea, hair loss, mouth sores, a nasty taste in your mouth and fatigue (lack of energy). But don't worry, these are all temporary and will stop when the chemo is finished. Medicines can be given before and after chemotherapy to help with some of these side-effects, such as anti-sickness drugs.

The best people to talk to about side-effects and how to cope with them are doctors, nurses and other young people who have gone through treatments for cancer.

DEALING WITH SIDE-EFFECTS

Each cancer patient is different and when doctors tell you the list of side-effects that you could experience, remember that you might not get them at all, you might only get some of them, or you might not get the side-effects as severely as other patients.

Don't worry about this, it doesn't mean your treatment isn't working if you don't have the same side-effects as someone else on the same treatment.

Chemotherapy works by being toxic to rapidly dividing cancer cells!

NAUSEA AND VOMITING

The doctors can give you anti-sickness tablets so this should not be too much of a problem. Try to avoid very strong flavoured food or fried greasy food. Eating small regular meals will help. Distracting yourself by watching TV whilst eating can take your mind off any sickness you may start to feel.

NEUTROPENIA

Before and during your treatment you will have regular blood tests to check levels of special blood cells called neutrophils (sometimes called 'neuts'). Neutrophils are white blood cells that are an important defence against bacterial infections. Chemo can lower the number of neutrophils in the blood, causing a condition called neutropenia, and this increases your risk of getting infections.

It is very important that you know what to do if you get a fever while you are receiving chemotherapy, as fever can be a sign of infection. Your MDT will give you a contact number at the hospital for you to use if you get a temperature. If this happens it is very likely that you will be asked to go to the hospital to get a blood test and to be seen by the oncology doctors. You will be given antibiotics to fight off an infection if you are found to have a neutropenic fever. Don't worry about mixing with other people – it's rare for people to pass infections on to you; try to continue with normal life as much as possible.

FATIGUE (LACK OF ENERGY)

Fatigue is the most common side-effect of chemotherapy and other cancer treatments. Fatigue is slightly different from tiredness. We all get tired but fatigue is a feeling of lacking energy when we shouldn't really be, and it doesn't seem to go even after a long sleep. Fatigue can also be caused by having too few red blood cells, this is another side-effect called anaemia.

There is a medication that doctors can prescribe to help with fatigue. You may have a blood transfusion if your fatigue is caused by anaemia. Research has shown that exercise can help to reduce the feeling of fatigue, and that resting all the time makes it worse. Maybe you could go for a walk or go shopping with friends, or find another way to stay active during chemo and the feeling of fatigue should improve.

SORE MOUTH

Mouth ulcers can be a side-effect of chemo; this can make it difficult or unpleasant to eat. Again, doctors can give you mouthwashes and medicated gels that can help. Brushing your teeth regularly with a soft toothbrush will also help to keep your mouth clean. Sucking ice cubes or eating ice cream while you are having the chemo may also help.

TASTE CHANGES

You may find that you have a taste of metal in your mouth or that your favourite foods suddenly taste horrible, salty or bitter. Don't worry - this will stop once the chemo is over.

Using plastic spoons, knives and forks may help to reduce the metallic taste. Adding herbs and spices to food and avoiding food that doesn't have much flavour may also help. Getting advice from a dietician is also a good idea.

Notes on Chemo

- Chemotherapy, often called chemo for short, is the name for drugs used in the treatment of cancer. These drugs kill cancer cells or stop their growth by interfering with the way cells divide and grow (also known as the cell cycle), or by damaging the cell's genes.
- Chemo is used as part of the treatment for osteosarcoma.
- Chemo works by being toxic to rapidly dividing cancer cells.
- Different chemo drugs affect different parts of cells and that is why more than one drug may be given. This is called combination chemo.
- Some of the body's healthy cells also divide rapidly like the cells lining the mouth, stomach and hair follicles. The chemo is also toxic to these cells and this is what produces side-effects.
- Chemotherapy for osteosarcoma is given directly into the blood by an IV line, or sometimes by tablet.
- Chemotherapy is given in 'cycles'. A 3 week cycle could include 3-4 days of receiving the drug and 2 and a bit weeks for the body to recover before the next cycle starts.

HAIR LOSS

This is the side-effect of chemo that most people know and ask about. Hair loss can begin from a few days to a few weeks after starting chemo.

People cope differently with hair loss, some may not be so bothered about losing their hair - others may feel upset, scared or distressed about it. Try to stay positive; the main thing to think about is that it will grow back. You may lose some or all of your hair, even eyebrows and pubic hair, this can be especially distressing. Talking to your nurse can help a lot

Options? You may feel you want to wear a head scarf, cap, beanie or a wig, or you may just want to bare all by shaving your head before the hair loss starts! You could even raise some money for charity or your hospital by being sponsored for your head shave.

TEENAGE CANCER TRUST RUN A 'HAIR 4 U' PROJECT TO ENSURE EVERY YOUNG PERSON WITH CANCER HAS ACCESS TO A REAL HAIR WIG. CONTACT HELLO@TEENAGECANCERTRUST.ORG TO FIND OUT MORE.

KEEPING YOUR WEIGHT STABLE

During your treatment you may lose your appetite and begin to lose weight. It's important to keep your weight as stable as you can, calories are really important at this time. Try to eat a nutritious diet that includes a variety of foods so that you get all of the energy, protein, vitamins and minerals you need to support your body and keep your immune system in good working order.

Children's Cancer and Leukaemia Group (CCLG) produce a really helpful booklet called 'Helping Your Child to Eat – a practical guide for Parents, Carers and Children Coping with Cancer'. This is full of helpful information about the best things to eat, and how to get all of the nutrition you need.

Visit their website at www.cclg.org.uk.

DO WHAT WORKS FOR YOU!

The take home messages for dealing with side-effects are; everybody is different, get as many tips as you can from other patients, tell your Mum, Dad and carers about them and 'do what works for you!'

Megan's Story

Visit the Bone Cancer Research Trust website to read how former bone cancer patient, Megan Blunt dealt with eating during chemo...

www.bcrt.org.uk/patientstories

Megan also wrote a book, 'Chemotherapy, Cakes and Cancer' to provide an A-Z guide on dealing with treatment.

a nutritious DIET is really IMPORTANT

CHEMOTHERAPY IS GIVEN before & AFTER TREATMENT

HOW WILL I BE GIVEN CHEMO? WILL I NEED LOTS OF NEEDLES?

There are different ways you could be given chemo: tablets, liquid medicine, injection or directly into the blood.

Most of the time you will have chemo directly into your blood. The drug is given through a cannula (sometimes called a Venflon), which is a flexible thin plastic tube that sits in a vein in the arm or hand. Often patients have a central line, or Hickman line, 'PICC' or 'implantable port' (Portacath®).

Portacaths®, PICCs and central lines can be kept in for weeks or even a few months. These lines enable the number of needles required during treatment to be minimised and more than one drug or treatment (such as fluids or nutrition) can be given at the same time because the lines can have multiple openings.

Because central lines, PICCs and Portacaths® are all slightly different, the decision on which type of line will best suit you can be discussed with the nurses and the doctor.

The chemotherapy drug enters the blood through the cannula by Intravenous (I.V. for short) infusion, which is usually called a 'drip'. An infusion or drip is a method of giving a set amount (dose) of I.V. medicines such as chemo over a set period of hours or days. The medicine goes straight into a vein via a long tube that is connected to a bag that holds the chemo medicine. The medicine 'drips' into the blood slowly.

The infusion can also be controlled by an infusion pump, which is connected to a central line or a PICC line. Some of the pumps are small enough to fit in a pocket meaning that you can use them at home.

Chemotherapy is given in 'cycles.' A cycle is the treatment time plus a resting time. For example, you may be given a combination of chemo drugs over 3-4 days and then there may be a resting period of 2 1/2 weeks. This means the cycle is 3 weeks long. The resting period helps the healthy cells recover before the next treatment cycle begins.

WHAT IS MIFAMURTIDE (MEPACT)?

Mifamurtide (Mepact) is an osteosarcoma drug that is given after surgery and chemotherapy. Mifamurtide can only be given to patients whose osteosarcoma has not spread (metastasised) and who are between 2 and 30 years old.

Mifamurtide is given to patients twice per week for 12 weeks, then once per week for 24 weeks. It can be given at outpatient appointments so you don't have to stay in hospital the whole time.

Mifamurtide works by stimulating the immune system to attack the osteosarcoma cells. It has very mild side-effects - some patients feel shivery when they take Mifamurtide, but this goes away quickly.

CLINICAL TRIALS

The doctors may also talk to you about clinical trials. Clinical trials are ways that scientists and doctors can find out what the best ways to treat osteosarcomas are. Clinical trials are a crucial stage in developing a new treatment or improving a diagnostic test. These trials can involve different things; from answering a few questions in a survey, to trying out a new drug or treatment to test whether it is better than the usual treatment.

Before a clinical trial, the treatment or test under study will have been investigated in the laboratory during tightly controlled studies, known as 'preclinical studies', and may well have been in earlier clinical trials too.

TAKING PART IN A CLINICAL TRIAL

A doctor may suggest a clinical trial to you, or you and your family may find one online yourselves. Either way, it is very important that you speak with your doctor first to ensure you are able to take part and to ask any questions you may have before consenting to join.
If you are considering taking part in a clinical trial we would encourage you to use the information on Bone Cancer Research Trust's website to discuss your options and the clinical trials available to you with your doctor.

YOU ARE ABLE TO WITHDRAW FROM A CLINICAL TRIAL AT ANY POINT IF YOU WISH TO DO SO, AND THIS WILL HAVE NO EFFECT ON THE CARE AND FURTHER TREATMENT YOU WILL RECEIVE.

PROS AND CONS

There are many things to think about when considering taking part in a clinical trial. These include the need to travel to the centre where the trial is being held or the need to join a waiting list for some trials.

The treatment you receive during a clinical trial may or may not be better than the usual treatment; this is what the trial is trying to find out. However, during clinical trials patients are monitored closely and this often means that any changes to their health are picked up sooner and they may see better results.

This also means patients on a clinical trial will require more check-ups and tests than usual, and need to visit the hospital more often. This can be difficult for some patients but knowing that by taking part they may be helping other patients in the future can be a great bonus and a positive reason for lots of people.

BONE CANCER RESEARCH TRUST
UNTIL THERE'S A CURE

The Bone Cancer Research Trust provides in-depth information on clinical trials and what it means to take part in a trial. You can find this on our website at **www.bcrt.org.uk/clinicaltrials**

Alongside this we provide a list of the clinical trials that are available for primary bone cancer patients.
Get in touch with us if you'd like to know more at **info@bcrt.org.uk** or by calling **0113 258 5934**

BODY IMAGE

CANCER HAS CHANGED MY BODY – I DON'T KNOW HOW I FEEL ABOUT THIS

Cancer and cancer treatment can change your body. These changes in the way your body looks, feels and works can affect your body image – this means the way you now see your body and feel about your appearance.

WHAT ARE THE COMMON PHYSICAL CHANGES?

Changes in your body can be caused by cancer itself or by cancer treatments; including surgery, chemotherapy and radiotherapy. Some physical changes affect your appearance, whereas others affect how your body feels or functions. Some changes may be temporary, such as hair loss; others may be permanent. Common physical changes associated with cancer and cancer treatments include the following:

- **Scars from surgery**
- **A loss of part of your body or part of an organ in your body**
- **Loss of sensation (numbness)**
- **Weight loss or weight gain**
- **Hair loss**
- **Fatigue; this can cause changes in your physical abilities, such as feeling weaker, having less energy, or being unable to take part in the physical activities you used to enjoy**
- **Changes to how you feel about sex; this could be a loss of desire for sexual activity or physical changes that makes sex difficult.**
- **Changes to your reproductive organs, such as infertility (the inability to have a baby). Not all chemotherapy drugs have effects on fertility but if you are told that your treatment may cause infertility then female patients may want to think about and discuss with doctors the possibility of freezing eggs or ovarian tissue. Male patients could discuss the possibility of sperm banking.**

HOW WILL PHYSICAL CHANGES AFFECT HOW I FEEL ABOUT MY BODY?

Visible changes, such as skin problems or hair loss, directly affect body image by changing the way you look physically. Changes like these may make you feel self-conscious about your appearance. Changes that aren't easily visible, such as a hidden scar, can also affect the way you feel about yourself. Even if you don't look different to others, you may feel insecure about your body.

Talking to others is a great way of finding out how they dealt with a similar situation to yours

REMEMBER,

AREN'T CHANGED BY CANCER

CONTACT THE BONE CANCER RESEARCH TRUST FOR HELP GETTING IN TOUCH WITH OTHER PEOPLE AFFECTED BY BONE CANCER.

You may feel that others see you differently because you have cancer, even if you look the same. Not all changes are negative, though. Some people have positive changes in body image, such as appreciating the strength of their bodies or discovering that issues such as weight or body shape matter less.

DO YOU HAVE ANY TIPS ON HOW TO COPE WITH HOW I FEEL ABOUT THE CHANGES TO MY BODY?

The tips below can help you cope with body image concerns. Remember that it's natural to feel anxious or worried about changes to your body. Give yourself time to get to grips with any physical changes and to adjust to how you feel about your body now:

- **Talk to others who are going through or have been through a similar situation to yours. It'll help to hear how they coped with changes to their body.**
- **Be prepared for questions and comments about the way you look and practice with a friend how you'll respond.**
- **If people do comment, try not to take it too personally. Remember that people are not trying to hurt you on purpose; they may feel awkward or uncomfortable and not know what to say.**
- **Even though your body may look and feel different, remember you are still the same person on the inside. Your personality, interests, and talents aren't changed by cancer.**
- **Try out new looks – experimenting with makeup, wigs and headscarves could give you the extra confidence you need.**
- **Take care of your body with good food, lots of sleep and an exercise plan approved by your doctor.**
- **If you're unable to do some of the activities or sports you used to, try to find a new activity that you'd enjoy. Learning a new skill can help you regain confidence in yourself and your body.**

AFTER TREATMENT: HOW WILL PRIMARY BONE CANCER AFFECT ME IN THE FUTURE?

When a patient has an illness, doctors try to predict how that illness will affect the patient, for example how likely it is that the treatment will work, and how likely it is that the patient will be cured. This is called a 'prognosis'. Lots of different things are taken into account when doctors make a prognosis for osteosarcoma. The main ones are:

- **The site of the tumour.** The position of the tumour in the body can affect your treatment.
- **The size of the tumour**
- **The stage of the cancer.** Whether the cancer has spread (metastasised) at the time of diagnosis.
- **The age of the patient and the patient's general health.**

Again, the thing to remember here is that each patient is different and doctors can never be certain how osteosarcoma will affect you.

AFTER TREATMENT

WHAT NOW?

The end of your treatment will come as a welcome relief but it can also be a time of worry and anxiety. You may be worried about relapse or your cancer coming back. You may feel lost and alone or be worried about leaving behind the hospital, the huge medical team and the support network that you have been used to. Your parents or carers may also be experiencing these emotions. Again, these are all perfectly normal feelings. You may like to join a support group or contact other primary bone cancer patients through blogs or Facebook.

BONE CANCER RESEARCH TRUST CAN PUT YOU IN TOUCH WITH OTHER PATIENTS. VISIT **WWW.BCRT.ORG.UK/SUPPORT** OR CALL US ON **0113 258 5934**

Other organisations may also be helpful for you:

- **JTV Cancer Support**
  www.jtvcancersupport.com
- **Teenage Cancer Trust**
  www.teenagecancertrust.org
- **CLIC Sargent**
  www.clicsargent.org.uk
- **Children's Cancer and Leukaemia Group (CCLG)**
  www.CCLG.org.uk
- **Sarcoma UK**
  www.sarcoma.org.uk

FOLLOW-UP CARE

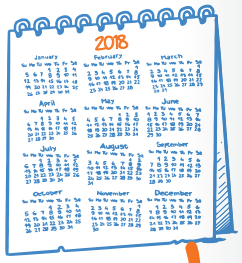
When you finish treatment your doctors will want to keep an eye on you and you will be monitored for a long time. This is called 'follow-up care.'

Outpatient hospital visits will be needed on a regular basis, for example; every 2 months in the first year, every 3 months in the second year, every 4 months in the third year and then 6 monthly until 5 years. After that, follow-up appointments are likely to be annually.

These visits will help to keep an eye on your general health as well as an opportunity to carry out some tests. These tests are very important because they can show up any 'late-effects' from the cancer treatment and to check for any relapse or recurrence of the cancer. If you have any problems in between these appointments you can get in touch with your centre straight away.

Follow-up care with an orthopaedic surgeon also helps to look out for surgery-related complications and to make sure the limb is working well. These appointments can also be useful to bring up any emotional or practical issues you may have.

Try not to let your cancer define your life, try to have medium & long term GOALS!



During the time between your outpatient appointments, you may feel unable to plan for the future; one year, or even 3 months ahead. But try to not let cancer define your life; try to have medium and long term goals.

The best way forward is to plan ahead; plan to do your A levels, plan to get into college or university or go on a back packing trip to Australia in 18 months.

BE POSITIVE, LIVE YOUR LIFE!

WHAT IF THE OSTEOSARCOMA COMES BACK?

Unfortunately, osteosarcoma does come back in some people. If your cancer does return, it may feel like the bottom has fallen out of your world, but there are treatments that you can be given.

You will probably be given a different combination of chemo drugs. The treatment may be more aggressive than your first treatments, meaning that higher doses are used. You may need further surgery and possibly radiotherapy. However all cases are different and treatments will be individual to each patient.

The doctors and nurses will be able to talk you through the treatment options in detail.

REMEMBER, YOU'VE GOT THROUGH THIS ONCE!

ADVANCED OSTEOSARCOMA

Your doctor may have told you that your treatment is no longer aimed at curing your cancer, but to try to prolong your life or make you comfortable.

You can find more information on this and coping with your emotions by contacting the Bone Cancer Research Trust support givers, at **www.bcrt.org.uk/support**, or by visiting the following pages:

- **Macmillan Cancer Support**
  www.macmillan.org.uk/cancerinformation/teensandyoungadults/lifeaftercancertreatment/
- **CLIC Sargent**
  www.clicsargent.org.uk/content/cancer-information-young-people
- **Teenage Cancer Trust**
  www.teenagecancertrust.org/get-help/ive-got-cancer/treatment/after-treatment

GLOSSARY

You might hear a lot of medical words and terms that don't make sense at first when you talk about osteosarcoma. Here is a list of some of these words, with explanations about what they mean.

A

Anaemia Lower than normal levels of red blood cells or haemoglobin in the blood.

Analgesic Drugs (medicines) that help to stop patients feeling pain. Examples include ibuprofen, paracetamol and codeine.

Antibiotics Drugs that fight infections.

Antiemetics Drugs that stop patients from feeling sick (nausea) or being sick (vomiting).

Autograft Replaces tissue from one part of the body using tissue from another area of the patients body.

B

Benign A mass of cells that is not cancerous. These tumours grow in one place and do not spread to neighbouring tissues or distant organs

Biopsy A biopsy is a small sample of a tumour that is taken so a doctor called a pathologist can look closely at some of a patient's cells or a piece of tissue and run tests on the sample.

C

Cell All living things are made up of cells. They are the building blocks of the body. Cells are very, very small and can usually only be seen under a microscope.

Central Line Long, flexible, plastic tubes that go into a 'central' blood vessel in the chest near the heart. These are used to give chemo drugs.

Chemotherapy (Chemo) The treatment of cancer with medicines that stop the growth of, or kill cancer cells.

Cisplatin Cisplatin is a platinum-containing chemotherapy drug that is given as a treatment for osteosarcoma. This drug works by binding or sticking to DNA and starting a process called apoptosis, which will kill the cell.

Clinical Trial Clinical trials are used to test how well new drugs or treatments work in people.

Counsellor A person who provides counselling, which is a type of talking therapy that allows a person to discuss their problems and feelings in a confidential and dependable environment.

CRP 'C-Reactive Protein;' blood tests to measure the levels of CRP are carried out to look for signs of inflammation and infection.

Curettage A method of surgery which removes the tumour by scraping or scooping the cancerous cells away.

Cyclophosphamide A chemotherapy drug used to treat primary bone cancers. The drug belongs to a class of drugs called alkylating agents. Cyclophosphamide slows the growth of cancer cells by interfering with the actions of DNA.

Cytotoxic Toxic to cells; a cytotoxic drug is a cell-killing drug.

D

Dexamethasone A steroid drug. Dexamethasone is a type of drug known as a corticosteroid. The drug has an anti-inflammatory effect and is used to decrease inflammation. Dexamethasone may be used to treat some of the side-effects caused by chemotherapy.

Diaphysis Also called the shaft, is the main part of a bone between the epiphyses.

Diclofenac A drug used as a painkiller, which belongs to a group of drugs known as non-steroidal anti-inflammatory drugs (NSAIDs). Diclofenac may be used to treat pain following an operation.

Distal Situated or placed farthest from point of

attachment to the body or origin. For example, the femur (thigh bone) is attached to the hip; (its point of origin) so the distal part of the femur is the part of the bone furthest from the hip (nearest to the knee joint). The opposite of distal is proximal.

DNA DNA stands for deoxyribonucleic acid. DNA is a long molecule (as long as 2 metres) packaged into the centre of each cell. DNA is the genetic material of a cell and contains genes or 'instructions' that are passed on from one cell to the next when they grow and divide.

Doxorubicin A chemotherapy drug that is given as a treatment for primary bone cancer. Doxorubicin is a red fluid and is given directly into a vein by infusion. It is a type of drug called an anti-tumour antibiotic. It works by interfering with the cancer cell's DNA at a time when the cell is getting ready to divide.

E

ECG A test that measures the electrical activity of the heart. This test is usually carried out before and during treatment to see how well the heart is working and detect any damage to the heart at an early stage.

Echocardiogram This test is often referred to as an 'Echo' and it is used to examine how well the heart is working.

Enneking System A staging system of musculoskeletal tumours to classify and describe tumours in order to study the extent of the cancer

Epiphysis The growth areas near the ends of a bone.

Etoposide A chemotherapy drug that is given as a treatment for primary bone cancer. The drug is in the form of a white powder, which is then made into a colourless solution. Etoposide is usually given directly into a vein by infusion.

F

FBC(Full Blood Count): A blood test which gives information about blood cells and how many cells there are. It looks at: Red blood cells, white blood cells (totals of each type), platelets and levels of a substance called haemoglobin, which carries oxygen in the blood.

Febrile Neutropenia Fever (higher than normal temperature, which is a sign of infection) and neutropenia, which is a lower than normal number of a type of white blood cells in the blood called neutrophils.

Femur The thigh bone, the longest bone in the body.

G

Gene Genes are the instructions that contribute to the function of a cell. The information for these instructions is stored as a code on a long molecule called DNA. Humans have around 20,000 genes.

GFR Glomerular filtration rate (GFR) is a measure of how well a patient's kidneys are working.

Grade The grade of a cancer describes how active the cancer is. Tumours are graded by pathologists who look at tumour samples under the microscope. One of the things they look at is how many dividing cells they can see. High grade tumours are often more aggressive.

H

Hickman® Line A type of central line used to give fluids, nutrition, chemotherapy and other drugs directly into the blood, and to take blood samples.

High- Dose Chemotherapy Intensive cancer treatment using very high doses of chemotherapy drugs to kill cancer cells.

I

Ifosfamide A chemotherapy drug that is given as a treatment for primary bone cancer. Ifosfamide is given directly into a vein by infusion (drip) or injected into a vein using a cannula. Ifosfamide is classed as an alkylating agent and works by damaging DNA and preventing it copying itself, interfering with cell division.

Iliac Around the area or region of the ilium (largest bone of the pelvis), for example, the iliac artery.

Infusion (drip) An infusion or drip is a method of giving a set amount (dose) of I.V. medications such as chemotherapy or painkillers over a set period of time. This period can be hours or days.

Investigation Tests carried out by doctors to aid in the diagnosis and monitoring of illness.

Intravenous (I.V.) Giving medicines or fluids directly into the blood through a tube called a cannula or Venflon, which sits inside a vein, usually in the arm or back of the hand.

L

Lactulose A medicine used to treat constipation (difficulty with bowel movements). Medicines used to treat constipation are called laxatives.

Late-Effects Problems that patients can sometimes develop after cancer treatments such as chemotherapy have finished (sometimes months after).

LFTs Liver Function Tests; carried out using a blood sample to examine the levels of certain substances released by the liver. These tests help to show how well the liver is working.

Limb-Sparing Surgery Also called Limb Salvage Surgery. This type of complicated surgery is carried out to remove bone cancers, avoid amputation and to preserve the patient's appearance. It also helps to keep as much of the limb function as possible.

M

Malignant A malignant tumour or cancer is a 'lump' or 'growth' of tissue made up from cancer cells, which continue to multiply and are able to move in the blood or lymphatic system to other parts of the body where they can cause damage.

Metastasis Malignant cancer cells may break off from the primary (original) tumour and travel to new sites in the body in blood or lymph vessels where they can form new tumours. This means the cancer has spread.

Methotrexate A chemotherapy drug that is given as a treatment for primary bone cancers. Methotrexate is given directly into a vein by infusion or injected into a vein using a cannula. The drug is in a class of medications called antimetabolites. Methotrexate treats cancer by slowing the growth of cancer cells; the drug interferes with the production, or more accurately the synthesis, of DNA.

Metoclopramide (Maxolon) A drug used to treat nausea and vomiting. The drug works by speeding up stomach emptying by increasing the contractions of the stomach and intestines.

Morphine Sulphate A painkilling drug used to treat severe pain. Morphine sulphate belongs to a group of drugs called opioids. The drug is taken in the form of swallowing tablets or a syrup.

Mifamurtide (Mepact) Mifamurtide is also reffered to as its brand name, Mepact. It is a treatment for osteosarcoma patients between 2 and 30 years. Mifamurtide is given alongside chemotherapy after surgery to lower the risk of the osteosarcoma coming back.

Multidisciplinary Team The Multidisciplinary Team (MDT) is made up of people who are experts in different areas of medicine. They work together following national guidelines to work out patient treatment plans, decide on appropriate further tests and referrals, and keep records.

N

Nasogastric Tube A thin flexible tube which is inserted into the nose and down the back of the throat into the stomach. Nasogastric tubes can be used in two ways; to provide nutrition or medicines or to remove the contents of the stomach, including air.

Neoplasm A neoplasm is another name for a tumour. The word neoplasm means 'new growth'. Neoplasms are either benign (not cancer), or malignant (cancer).

Neutropenia A low count of neutrophils. Chemotherapy can reduce on the number of neutrophils in the blood. This lowers the immune system, which helps to fight off diseases. If a patient has neutropenia they will be at a greater risk of infection and may be given antibiotics to prevent an infection or to treat an existing infection.

Neutrophils or 'Neuts' White blood cells that are part of the immune system, which helps to fight off disease. Neutrophils are the most common type of white blood cell. They are a type of cell called a phagocyte, which means they 'eat' bacterial cells that may be infecting the body.

O

Occupational Therapist Occupational therapists help people who have been ill or undergone surgery to adapt to changes caused by their illness. This might include helping the patient to get around in day-to-day life or return to work/school.

Oncologist A doctor who treats and looks after people with cancer.

Ondansetron A drug used to treat or prevent nausea and vomiting associated with cancer treatments and surgery.

Orthopaedic Surgeon Orthopaedic surgeons treat conditions using surgery that affect the bones, joints, muscles, ligaments, tendons and nerves.

Osteoblast Cells that specialise in making bone. Bone is not just made during development and childhood; it is constantly remodelled or reshaped throughout life. Osteoblasts work in close balance with osteoclasts.

Ostoeclasts Cells that breakdown bone and

osteoblasts rebuild the bone. Osteoclasts work in close balance with osteoblasts.

Osteosarcoma A malignant bone tumour that often develops when bones are growing quickly. That is why most osteosarcomas are found in older children and teenagers. It is the most common primary bone cancer in children, teenagers and young adults.

Outcome The result of a disease that directly affects the length or quality of a person's life.

P

Paediatrician A doctor who specialises in treating children.

Palliative Medicine that helps to support patients and to manage or relieve their symptoms to improve their quality of life.

Pathologist A doctor that uses laboratory techniques to diagnose disease.

PEG Percutaneous Endoscopic Gastrostomy. It is used to help keep a patient's weight stable if they are unable to eat or swallow. A PEG is used to get nutrients and fluids directly into the stomach.

Physiotherapist A physiotherapist (sometimes shortened to physio) will assist with rehabilitation after surgery.

PICC Peripherally Inserted Central Catheter. It is a long thin tube that is inserted into a vein in the arm, usually near the elbow, and then fed through the vein until the end or tip reaches a vein near the heart.

Portacath® Also called implantable ports, consist of a thin tube (catheter) and a port or chamber. The tube is implanted under the skin in the chest and sits in a vein near to the heart. The port is attached to the end of the tube and sits just under the skin. The port contains a thin rubber disc through which special needles can be attached to give chemotherapy and other I.V. treatments, fluids and to take blood samples.

Prednisolone Prednisolone is a type of drug known as a corticosteroid. The drug has an anti-inflammatory effect and is used to decrease inflammation. Prednisolone may be used to treat some types of cancer, some infections and some of the side-effects caused by chemotherapy.
Prognosis A prediction of the probable course and outcome of a disease.

Promethazine A drug used to treat allergic reactions or as a sedative (to induce calm and sleepiness) before surgery and insomnia (inability to sleep). Promethazine belongs to a group of drugs called antihistamines.

Prophylactic 'Prevention'; a medication or treatment, which is used to prevent a disease or infection from occurring. For example, after surgery prophylactic antibiotics may be used to prevent infection of the wound.

Prosthesis A device designed to replace a missing part of the body.

Prosthetist A specialist who looks after anyone who requires an artificial limb. They can help to find the best prosthesis (artificial limb) for the patient and give advice on rehabilitation.

Proximal Situated or placed close to the point of attachment or midline of the body (centre of the trunk). For example the femur (thigh bone) is attached to the hip, (its point of origin) so the proximal femur is the part of the bone 'nearest' to the hip. The opposite is distal.

R

Radiotherapy Radiotherapy treats tumours using high doses of concentrated radiation. In some patients with primary bone cancer, radiotherapy is given as a single dose each day (for a few minutes) for approximately 5 - 6 weeks.

Randomised Clinical Trial To reduce any chance of bias, patients are put into random groups when taking part in a clinical trial. In its simplest form, patients will be randomly assigned by computer into one of two groups.

Recurrent/Relapsed Cancer This is cancer that has come back either to the same place as the first cancer or to another place in the body.

Regimen A plan of the medicines or other treatments patients may have. It lets all the carers know exactly how a patient will be treated. The regimen will say what sort of treatment the patient will receive, the dose, how often and how long it will be given for.

Rehabilitation Often shortened to 'rehab.' Rehab is a form of treatment which helps patients to do the things they did before they got ill, such as helping them to walk after surgery. Physiotherapists and Occupational Therapists are specialists in rehabilitation.

Remission When the signs and symptoms of cancer have disappeared, patients are said to be in remission. Being in remission may not mean they are cured.

Resection Surgical removal of tissue (such as bone) or an organ from the body.

S

Sarcoma A tumour that starts in connective tissue. The name sarcoma is thought to come from the Greek word 'sarx,' which means tissue. Bone sarcoma is another term for bone cancer.

Second Opinion Getting the view of a doctor who has not been previously involved in a patient's case. A second opinion is usually obtained by a referral from a GP. There is no legal right to a second opinion.

Serum Clear liquid that remains after blood cells and clotting proteins have been removed from a blood sample.

Shared Care Hospital Treatment for primary bone cancer could take place in a hospital quite a long way from where patients live. If there are problems between treatments, such as an infection, the patient may have a 'shared care hospital,' which is usually their local hospital.

Side-Effects Side-effects happen when drugs or medicines that patients are given to treat a disease also have an effect (usually unwanted) on parts of the body that are healthy.

Stage The stage of a cancer describes its size and if it has spread to other parts of the body.

Steroids Drugs (corticosteroids or glucocorticoids) which are similar to hormones produced by the body in response to stress, illness and injury. Steroids are used to reduce inflammation, and are used to treat some types of infection, some types of cancer or side-effects caused by cancer treatments

Systemic 'The whole body'. Chemotherapy for primary bone cancer is a 'systemic' treatment; it treats the whole body.

T

Tissue A group of cells that work together to do a certain type of job in the body.
Topotecan A chemotherapy drug that is given as a treatment for some types of cancer.
Topotecan works by stopping an enzyme called Topoisomerase I, resulting in damaged DNA, which causes the cell to die (by a process called apoptosis).

Total Parenteral Nutrition (TPN) Also called PN or Parenteral Nutrition. This is when a patient is given nutrients (fed) directly into the bloodstream though a drip.

Treatment Cycles A cycle includes the treatment time, which could be 3-4 days plus a rest period for the healthy cells of the body to recover. A typical cycle for primary bone cancer treatment could be three weeks.

U

Us & Es Urea and Electrolytes; a type of blood chemistry test to examine the levels of dissolved salts as well as urea and creatinine, which are waste products. This test can give clues to how well the kidneys are working.

V

Venflon A Venflon, also called a cannula, is a small flexible plastic tube that is inserted through the skin into a vein by a needle which is attached to it. The needle is withdrawn when the tube is inside the vein. Venflons are used to give medications, fluids, and blood products directly into the blood (I.V.).

W

White Blood Cells Blood cells that are part of the immune system. The immune system helps the body to fight off infections and disease. White blood cells are also known as leukocytes.

If you've heard or seen a word that we don't have in this glossary, please let us know by emailing info@bcrt.org.uk so we can add it in the future!

USEFUL ORGANISATIONS

BONE CANCER RESEARCH TRUST

The Bone Cancer Research Trust (BCRT) is dedicated to providing up-to-date, trustworthy information for patients and their families. On our website you'll find downloadable information, hints and tips and can read stories from other patients going through a similar situation. **www.bcrt.org.uk**

LINKS TO OTHER CANCER CHARITIES AND USEFUL ORGANISATIONS

CCLG (Childrens's Cancer and Leukaemia Group)

Have lots of really helpful information for young people with cancer and their families. **www.cclg.org.uk**

Chemotherapy, Cakes and Cancer

Former osteosarcoma patient Megan Blunt wrote this book to share her top tips for long hospital stays and staying positive. **www.clicsargent.org.uk/content/chemotherapy-cakes-and-cancer**

Citizens Advice Bureau

Advice on legal issues, employment issues, money issues and lots more. **www.citizensadvice.org.uk**

CLIC Sargent

Provide information and support for children and young people with cancer. **www.clicsargent.org.uk**

Department of Work and Pensions

For advice on benefits you or your parents may be entitled to. **www.dwp.gov.uk**

Irish Cancer Society

A national charity dedicated to preventing cancer, saving lives from cancer and improving the quality of life of those living with cancer. **www.cancer.ie**

JTV Cancer Support

JTV Cancer Support is a video project aimed at educating people about the effects of cancer; not just the medical effect, but also the emotional effects upon the patient and their friends, family or anyone surrounding them by sharing experiences. **www.jtvcancersupport.com**

Macmillan Cancer Care

Provide a wide range of information specifically for teenagers and young people as well as practical and financial support. **www.macmillan.org.uk**

SCAT Bone Cancer Trust

Offer knowledge and advice on the best technology available in the manufacture of artificial limbs, special crutches or wheelchairs. **www.scatbonecancertrust.org**

Teenage cancer Trust:

A charity devoted to improving the lives of teenagers and young adults with cancer. **www.teenagecancertrust.org**

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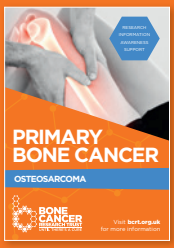
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